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Usability assessment and improvement strategy of music APP interface based on heuristic evaluation

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With the increasing diversification of virtual interface functionality and services, excessive cognitive and operational loads are posing new challenges to the usability of music app interfaces. This study investigates whether interfaces characterized by integrated functionality and complex operational processes effectively meet users' usability requirements. Through a heuristic evaluation methodology, 15 independent evaluators assessed the interface usability of the QQ Music App. The evaluation identified several critical usability issues, including inadequate visual hierarchy, insufficient distinctiveness in functional elements, excessive use of pop-up interfaces, and limited operational feedback mechanisms. This research indicates that interface usability and product competitiveness can be enhanced by using multiple strategic methods: optimizing the information load of the online music application interface, enhancing the visibility of common interface functions while simplifying operational processes, incorporating interactive prompts for popup management, and delivering interface feedback that corresponds with users' contextual engagement with the online music application. The effectiveness of these strategies was additionally corroborated through prototype creation and usability testing. This study's findings provide novel insights for design techniques applicable to music app development teams.

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Introduction

As we enter the digital age, individuals' daily lifestyles are evolving due to the widespread adoption of digital gadgets and mobile technology. Social services are becoming increasingly reliant on the Internet, and the utilization of mobile applications is expanding rapidly (Wu and Moody et al., 2020). Individuals today are more engaged in with daily activities through online applications than ever before. The mobile music industry has experienced tremendous growth in recent years, with digital platforms like Apple Music and Amazon Music emerging as the predominant channels for music distribution (Webster, 2019; Hracs and Webster, 2021). The online music experience has disrupted conventional methods of listening to music via records or compact discs (CDs), allowing individuals to access diverse genres and artists at any time and place, transcending traditional constraints of time and space, and significantly enhancing the efficiency and convenience of music distribution.

As of December 2022, the user base of online music applications in China has reached 710 million, indicating that over half of the Chinese population uses these platforms (Foresight Industry Research Institute, 2021). Simultaneously, online music applications are consistently enhancing the variety of interface functions and online service offerings while introducing a broader range of innovative interface features (Rahimi and Park, 2020). Figure 1 illustrates that the 2015 iteration of the QQ Music App interface preserved only fundamental functions, including music playback, music charts, and song categorization. This design enhanced user compatibility and reduced interaction burden. Nonetheless, contemporary computer interfaces are growing increasingly intricate to support additional functions (Miraz et al., 2021). Users' customization requirements have shaped today's 2024 release of the QQ Music App UI version, which offers enhanced features and online services such as a music library, children's music, audiobooks, live music streaming, and music collections. The increased information within the interface introduces new interaction complexities for users and generates additional usability concerns.

Thus, it is essential to prioritize the development of a "user-friendly" interface to enhance customer satisfaction with online music applications. Therefore, it is crucial to determine whether users face usability problems when interacting with the interface and to devise strategies for improving the interface's usability in light of the growing trend of integrating functional services into online music applications.

The purpose of this study is to understand the usability of online music APP interfaces in multifunctional and complex interface environments, and to accurately grasp the direction of usability improvement of online music APP interfaces by identifying the usability problems with a higher degree of severity among them. As of 2023, QQ Music App in China has attained a user base of 594 million, making it one of the largest online music applications currently available. Therefore, utilizing the QQ Music App for the case study provides a more accurate representation of the research subject. The study's conclusion provides valuable insights into the interface usability of online music applications.

This study employs heuristic evaluation, a widely recognized usability assessment method in human-computer interaction research (Cho et al., 2022), to examine the interface usability of the QQ Music application. Heuristic evaluation, through expert assessment, enables the identification of specific usability challenges. The research addresses three primary questions:

RQ1: How do users navigate and interact with complex interface systems in online music applications?

RQ2: What is the relationship between interface complexity and user experience in online music applications?

RQ3: How can we implement these improvements at a foundational level, and what opportunities exist for enhancing interface usability?

Six sections structure the study. The first section establishes the research context, rationale, objectives, and methodological framework. The second section presents a comprehensive literature review, examining existing research on online music applications and developing a theoretical framework based on usability principles and heuristic evaluation methodologies. The third section details the experimental design, procedures, and materials. The fourth section presents data analysis and research findings. The fifth section proposes specific interface improvements for online music applications, incorporating interface prototyping to iterate and enhance the QQ Music application. The final section synthesizes the research findings, discussing theoretical contributions and practical implications for online music application interface design while acknowledging research limitations and suggesting future directions.

Related works

Review of online music APP related research. This study conducted a comprehensive research assessment, concentrating on journal articles about the usability studies of online music applications published from 2018 to 2024. To ensure extensive literature coverage, searches were performed in the Google Scholar and Web of Science databases. Boolean operators (AND, OR) were utilized in the search strategy to improve the efficiency and quality of keyword retrieval and to identify literature pertinent to the research issue of online music app usability. For instance, standard search queries comprised "music app" AND "usability" or "online music mobile application" and "usability," which refined research publications that particularly address usability-related studies via keywords, abstracts, and titles. The research primarily analyzed conceptual pieces, case studies, literature reviews, empirical studies, and conference papers to obtain a variety of perspectives and ideas within the literature. Concerning the criteria for literature inclusion and exclusion, only papers written in English were deemed acceptable due to language limitations, and the following exclusion criteria were applied:

1. Articles unrelated to online music applications and usability evaluations.
2. Papers devoid of substantial content or scholarly rigor.
3. Articles disseminated in informal publications lacking rigorous academic standards or peer assessment.

The review of the literature shows that most of the latest academic research on music apps is focused on three main areas: the user experience of online music apps (Bouraqia et al., 2020; Choi et al., 2023; Qi and Cai, 2021; Hracs and Webster, 2021); the adoption intentions and service utilization of users (Hwang, Oh (2020); Wu et al., 2020; Patel et al., 2020; Camilleri and Falzon, 2021); and the concerning research on the usability of online music applications (Hai et al., 2021; Hung and Chiang, 2024; Venkova and Nkpoikanna, 2024; Damayanti and Christian, 2024). The subsequent representative studies exemplify these study trajectories.

Initially, investigations concerning user experience in digital music applications have produced substantial findings. Wang et al. (2018) assessed interface design via eye-tracking tests, demonstrating that user satisfaction was maximized with grid-layout interfaces characterized by high color saturation and bright backgrounds. Tian, Zhao (2020) examined TikTok's design through the lens of flow experience, proposing solutions to

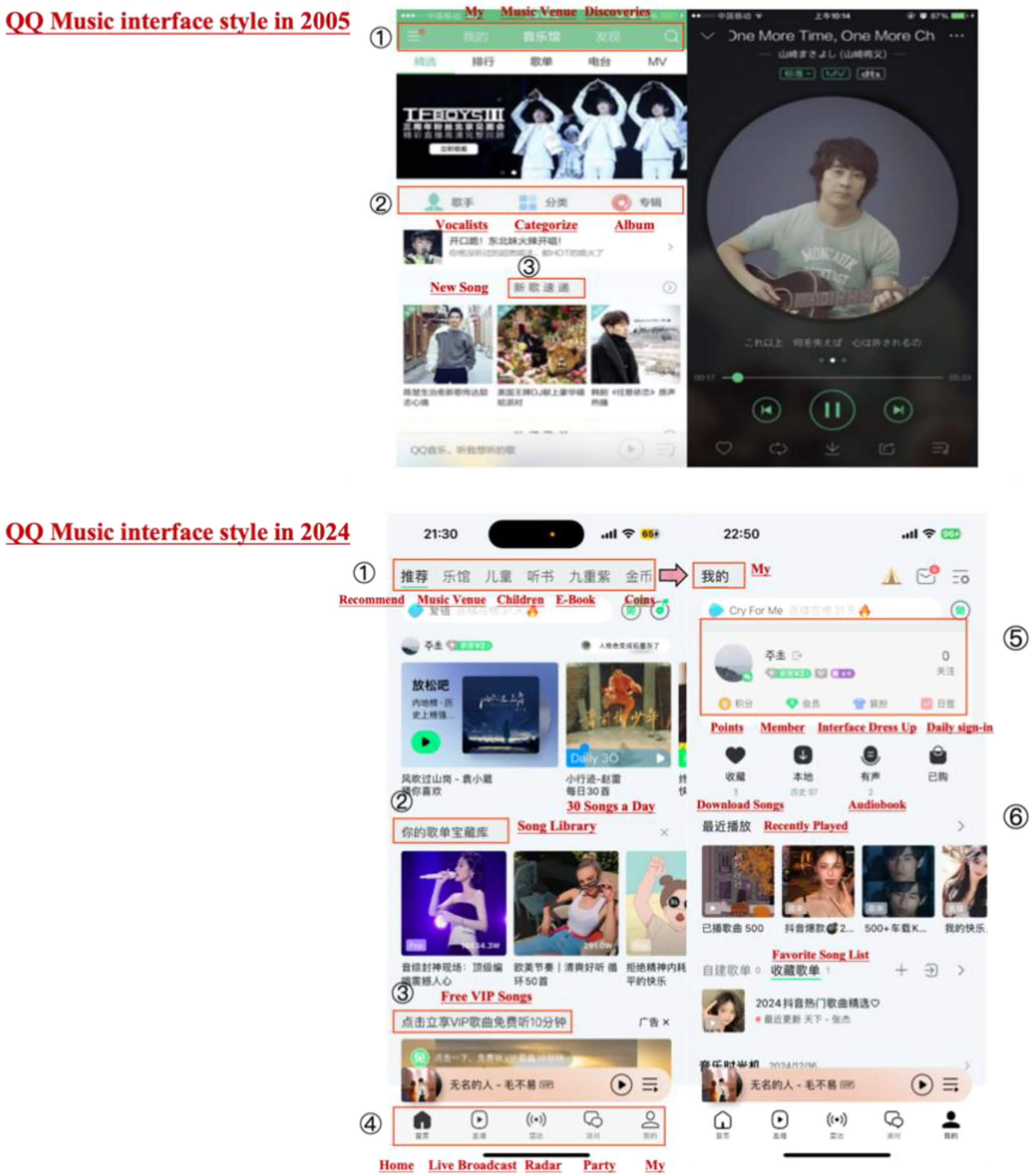

Fig. 1 Comparing the QQ music worlds of 2015 and 2024.

optimize product positioning, promote human-computer interaction equilibrium, and deliver prompt operational feedback. Hu (2019) assessed Chinese mobile music services through the lens of user experience, concluding that the three most popular services in China excelled in interface feedback, consistency, and memorability, whereas recommendation accuracy and interaction comprehensiveness necessitated enhancement. Fumić et al. (2023) identified that perceived ease of use and utility are essential factors determining satisfaction in music streaming services, with security and interface aesthetics significantly impacting perceived usefulness and interface aesthetics noticeably affecting perceived ease of use.

Secondly, research investigating the sustained usage intention of music applications has yielded significant insights. Camilleri et al. (2021) examined the motivations behind music streaming adoption through an integrated technology acceptance model and uses and gratification theory, demonstrating that perceived utility and ease of use are significant predictors of technology adoption intention, as users pursue emotional satisfaction via online streaming services. Zhang (2019) examined the influence of user experience elements on loyalty, revealing that visual presentation and emotional response improve perceived utility, which, in conjunction with platform reputation, favorably impacts user loyalty. Barata, Coelho (2021) examined the determinants of music streaming service purchases and recommendation intentions, indicating habit, performance expectancy, and price value as key factors driving the uptake of paid music streaming services.

Third, concerning research on the usability of online music applications, Hongjian and Yujie (2024) performed a usability study on the Spotify music application. The results showed that users preferred simple micro-interactions and perceived fewer, easier-to-understand interaction steps are better for reducing information and function overload. Users reported a preference for improved usability via greater customization and voice interaction techniques. Wang and Jia (2023) conducted a comparative usability analysis of interface systems among three music applications. Their research suggested that dark mode interfaces in music applications can enhance user operating efficiency. They highlighted the significance of evaluating the influence of conventional interface design layouts on user behavior throughout the implementation of interface enhancements. Carlsson (2024) investigated podcast app interaction patterns among Swedish listeners and identified the need for enhanced personalized discovery features and simplified user interfaces. Yu and Gao (2020) examined music app launch screens. Their research concluded that when designing music app launch screens, it is necessary to appropriately increase the white space ratio and pay attention to detailed text design. Chun and Park (2021) analyzed the interface usability of four music apps—Melon, YouTube Music, Genie, and FLO. Their research found that some apps' interface icons failed to adequately represent their functions. Additionally, insufficient visual feedback made it difficult for users to confirm successful interaction after clicking interface elements.

Recent research has identified the factors that influence the user experience with online music applications, user retention intentions, and the design standards for these applications. However, to date, only a limited number of usability evaluation studies that specifically focus on the interface of online music applications. Given the escalating complexity of functional services in online music applications and the growing demands of user interaction with the interface, it is imperative to reassess the usability of these interfaces to better identify their issues. This study was conducted to investigate the interface usability of online music applications in China, addressing the gap in existing research and considering the trend of functional integration. This study takes the QQ Music App, which boasts a substantial user base in China, as a case study to do a thorough usability review of its interface through heuristic assessment, aiming to uncover potential enhancements for the interface. The findings of this study may introduce novel design considerations for future online music application interface designs.

Usability and interface usability. The user interface serves as a mechanism for the system to provide information and interact with users (Medium, 2018). Interface design, layout, information design, and interaction design are critical components in software design that directly influence usability and user experience (Wang et al., 2022). Poor interface design will result in interaction gaps between users and mobile applications, hinder information transmission, create excessive user pressure, and lead to user rejection (Saputra and Kania, 2022). In recent years, the growing complexity of mobile applications' functionalities and interactive behaviors has rendered the enhancement of user interface usability increasingly critical. The International Organization for Standardization defines usability as the effectiveness, efficiency, and satisfaction of particular users utilizing items in specific contexts to accomplish designated objectives (Al-Qora'n et al., 2022). Usability is a critical criterion for assessing the effectiveness of a user interface system (Malik et al., 2021; Guimaraes et al., 2021; Ferreira et al., 2020), and the success of a user interface is predominantly contingent upon its usability. Enhancing product usability can help determine whether the system enables consumers to operate it conveniently and efficiently (Issa and Isaias, 2022; Mehmood et al., 2023).

In interface design, it is essential to enhance research on interface usability to improve both the quality and user experience of the design. An interface that is both appealing and easy to use can significantly increase user satisfaction (Saputra and Kania, 2022). Usability affects how well users can use a virtual interface. It also influences their efficiency and subjective experience. If the layout is not clear, if the navigation is hard to follow, or if the steps are too complex, users may make mistakes. It is essential to evaluate the usability of various iterations of the interface. Developers can enhance design when they understand the challenges consumers encounter (Issa and Isaias, 2022). Low usability may result in people operating the system less efficiently. They may also commit more errors (Farrahi et al., 2019; Nabovati et al., 2014; Jaspers, 2009). In interface design usability evaluation, human-computer interaction expert Jakob Nielsen defines user interface usability as attributes such as learnability, efficiency, memorability, error prevention, and satisfaction (Farzandipour et al., 2021).

Principles of heuristic assessment. Heuristic evaluation is a technique for assessing the usability of user interface designs. A limited group of evaluators assesses the interface to ascertain its adherence to established usability guidelines. This is among the most widely used techniques for assessing interface usability (Xiong et al., 2020; Kumar et al., 2019). In contrast to other usability testing methodologies, heuristic evaluation necessitates only a limited number of participants to assess the user interface and identify usability issues (Khowaja et al., 2020; Othman et al., 2022). Previous research indicates that heuristic evaluation is a useful approach for identifying user interface design issues (Joshi et al., 2009). Heuristic evaluation has advantages of time efficiency and ease of execution compared to other usability evaluation approaches (Khowaja and Al-Thani, 2020; Luna-Perejon et al., 2019; Dutta, Dhar (2024)).

Jakob Nielsen's heuristic evaluation principles are widely recognized standards for assessing user interface usability (Aldekhyyel et al., 2021) and are extensively applied in usability testing of diverse mobile applications (Kumar and Goundar, 2019; Aldekhyyel et al., 2021; Ihsan et al., 2024; Jainari et al., 2022). Furthermore, mobile application interface usability assessments extensively utilize Jakob Nielsen's heuristic evaluation techniques, demonstrating their efficiency and ease of application (Müssener et al., 2020). Thus, this study employed the 10 principles of Jakob Nielsen's ten heuristic principles as the criteria for assessing interface usability. Jakob Nielsen's 10 heuristic evaluation concepts encompass ten design principles pertinent to usability, specifically the visibility of system state. Alignment between the system and reality. User autonomy and liberation, Uniformity and principles, Error mitigation, Recognition over recall, Adaptability and operational efficiency, Aesthetic and minimalist design facilitates user recognition, diagnosis, and recovery from faults—assistance and documentation are crucial (Nielsen, 1994).

Methods and materials

Heuristic evaluation process. Recent studies (Kumar et al., 2020; Khowaja and Al-Thani, 2020; Iranmanesh et al., 2021)

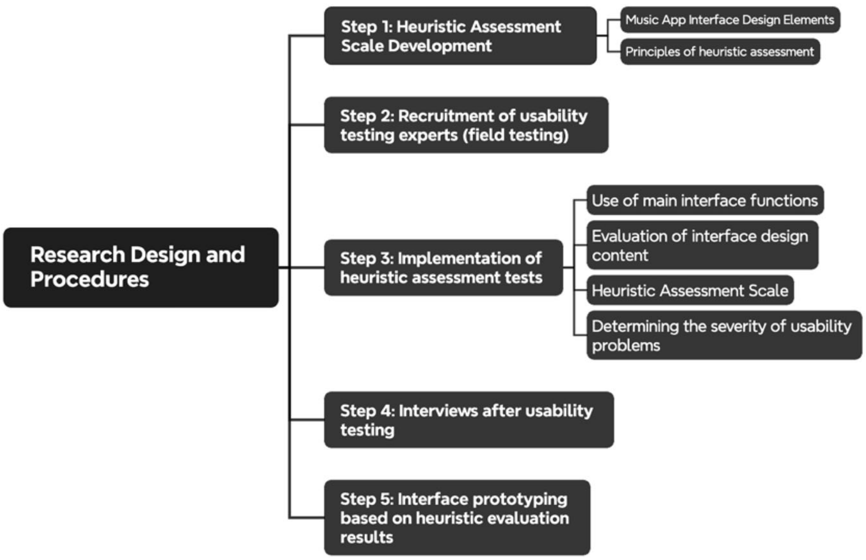

Fig. 2 Research Design and Process.

have extensively used heuristic evaluation to assess the interface design of diverse mobile applications, including mobile healthcare apps, mobile learning apps, and family health service apps. This heuristic evaluation study aims to assess the interface usability of China's QQ Music App, identifying usability concerns related to efficiency, effectiveness, and satisfaction, and therefore guiding the iterative design of future online music applications. As shown in Fig. 2, the research process of this study is as follows:

The study utilized an on-site testing methodology for the testing scenario. Participants were provided with a concise introduction to the heuristic evaluation through email, detailing the testing protocols and associated concerns, along with a reference checklist of interface testing components. During the formal testing procedure, participants were required to repeatedly scrutinize the interface, assess design components, and compare them with Jakob Nielsen's heuristic evaluation principles. The heuristic evaluation procedure unfolded as follows: To guarantee the quality of internal testing, researchers designated extra time slots for users to acclimate to the interface interaction patterns of the QQ Music App. Participants were instructed to allocate 30–50 min interacting with the app's core functionalities (music search, music recommendations, music playback, chart inquiries, etc.), recording operational feedback based on their actual user experience concerning interface navigation design, visual presentation, information architecture, usage flow, and other usability-related interface aspects. Subsequently, they finalized the heuristic evaluation scale and the usability severity scale presented in Appendix A.

Furthermore, to comprehend participants' evaluations of the QQ Music App's interface usability, each participant engaged in a 40–50 min semi-structured interview post-test, offering insight into the expert group's subjective viewpoints on the app's usability.

Development of the heuristic evaluation scale. Three usability experts developed the heuristic assessment questionnaire items relevant to this study, based on Jakob Nielsen's heuristic evaluation principles. They also referenced relevant research on questionnaire construction for mobile application heuristic evaluation (Tian et al., (2018)).

Table 1 Expert panel characteristics for delphi method study.

Characteristic	Category	N = 7	Percentage
Gender	Male	4	57.14%
	Female	3	42.86%
Professional role	Company UX Team Member	2	28.57%
	Company UI Design Specialist	4	57.14%
	University Faculty (UI/UX Domain)	1	14.29%
Educational background	Bachelor's Degree	1	14.29%
	Master's Degree	5	71.43%
	Doctoral Degree	1	14.29%
Years of experience	5–6 years	3	42.86%
	7–8 years	3	42.86%
	9–10 years	1	14.29%

The study adopted a 5-point Likert scale for gathering quantitative data. Seven experts in the UI/UX domain were consulted using the Delphi approach, ensuring the robust content validity of the questionnaire. Table 1 indicates that all seven specialists had 5–10 years of research experience in the UI/UX domain and exhibited substantial practical expertise in UI design creation, usability testing, and user research, thereby affirming their authoritative knowledge in the field.

To ensure the high reliability of the heuristic assessment scale produced and the efficacy of the testing process, three experts from the prior consultation round were chosen to review its content and perform preliminary testing.

Finally, following several rounds of expert comments, the research team enhanced the item descriptions, content, and readability of the evaluation scale. Table 2 presents the heuristic evaluation scale, which primarily draws from Jakob Nielsen's 10 principles of heuristic evaluation.

Participants and test equipment. This study adhered to established industry practices for identifying usability issues, drawing upon Nielsen's seminal research. This research demonstrated that conducting a usability evaluation by three to five experts can effectively identify approximately 75% of usability problems in a

Table 2 Heuristic evaluation scale for usability of online music app interfaces.

No	Nielsen's heuristics	Description	
1	Visibility of system status	1-1	Users should clearly understand their current operational status within the system
		1-2	Users should comprehend the implications represented by their operational behaviors
2	Match between the system and the real world	2-1	Users should understand the semantic meaning of interface functional icons
		2-2	The interface operation should align with users' operational habits
3	User control and freedom	3-1	Users should have freedom in selecting and canceling interface operations
		3-2	The system provides clear operational guidance to users
4	Consistency and standards	4-1	The interface layout, visual relationships, and navigation system maintain consistent design patterns
		4-2	The system's operational conventions maintain basic consistency with other related products in the industry
5	Error prevention	5-1	The interface provides clear prompts to prevent user operational errors
		5-2	When users make easily avoidable operational errors, the system provides appropriate confirmation mechanisms
6	Recognition rather than recall	6-1	The interface minimizes users' cognitive load regarding operational behaviors
		6-2	The interface provides prominent visual information to assist users in remembering operational procedures
7	Flexibility and efficiency of use	7-1	The interface provides efficient operational methods
		7-2	The interface allows users to perform repetitive operational actions
8	Aesthetic and minimalist design	8-1	While meeting user requirements, the interface maintains simplicity
		8-2	The interface can effectively display important operational functionalities
9	Help users recognise, diagnose and recover from errors	9-1	When users make operational errors, the system provides clear prompt information
		9-2	When users encounter operational errors, the system provides active solutions
10	Help and documentation	10-1	Users can easily access after-sales service interface assistance functions
		10-2	Help and customer service functions can effectively resolve user-encountered issues

Table 3 Basic information of UI/UX expert testers.

Project	Classification	Number of people ($N = 15$)	Percentage(%)
Sex	Male	9	60%
	Female	6	40%
Work experience or study experience	3-5 years	6	40.00%
	6–10 years	7	46.67%
	>10 years	2	13.33%
Occupation or education	UX Experiencer	6	40.00%
	UI Designer	4	26.67%
	Student	5	33.33%
Education level	Bachelor's degree	3	20.00%
	Master's degree	4	26.67%
	Master's degree students	2	13.33%
	PhD	6	40.00%

product (Nielsen, 1995; Adams and Hanson, 2020; Faulkner, 2003; Hettinger et al., 2023). To ensure methodological rigor and recruitment validity, participating experts were required to meet the following criteria: a minimum of three years of professional experience in UI/UX research or practice; substantial expertise in interface design; and extensive experience in applying heuristic evaluation methodologies.

Furthermore, we selected iOS as the uniform operating system for all test devices to maintain data collection and analysis consistency. This standardization was implemented to minimize potential confounding variables arising from system differences and enhanced the overall reliability of the data. All participants provided written informed consent before their involvement in the study.

Result

Participant characteristics. Between February 15 and February 29, 2024, the study enlisted usability experts via the Chinese social media platform WeChat employing the purposive sampling method. We recruited fifteen usability specialists to participate in the heuristic evaluation. Table 3 presents the demographic characteristics of the participants. To guarantee sample representativeness, these professionals possessed both extensive industry experience and profound scholarly backgrounds, together with proficiency in usability testing and evaluation.

Results of the heuristic assessment. A heuristic evaluation scale was used to assess the QQ Music App's interface usability under the test conditions illustrated in Fig. 3. The study collected 15 heuristic evaluation responses. To ensure the scientific rigor and validity of the data analysis and findings, SPSSPRO (https://www.spsspro.com/) was used to evaluate the scale's reliability and validity. While the limited sample size (15 evaluation responses) may affect the representativeness of the Cronbach's alpha coefficient, the reliability of the scale results can be supported by several factors: the participants' extensive experience in music app research and practice, and the rigorous review and validation process during scale development. The content validity of the scale was fully validated by expert review, further supporting its reliability. Additionally, post-test in-depth interviews were conducted to supplement the analysis. These interviews focused on understanding participants' scoring rationale, providing additional confirmation of the scale's reliability and consistency.

Figure 4 presents the results of the interface usability evaluation for the QQ Music App. Analysis of the mean scores from 15 evaluators yielded an overall interface usability score of 3.89, with a standard deviation of 1.52. Among the ten usability evaluation criteria, evaluators assigned relatively lower scores to system status visibility ($M = 3.63$, $SD = 1.12$), error diagnostics ($M = 3.75$, $SD = 0.81$), interface learnability ($M = 3.75$, $SD = 0.93$), and interface assistance ($M = 3.75$, $SD = 1.33$). It's worth noting that as this study involved a small-scale expert evaluation with a limited sample size (15 evaluation forms), some variability in standard deviations was expected. This variability, inherent to smaller sample sizes, does not compromise the validity of our findings.

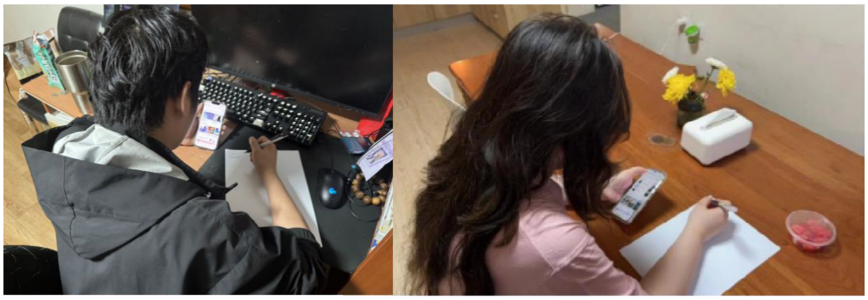

Fig. 3 Offline testing scenarios for heuristic evaluation.

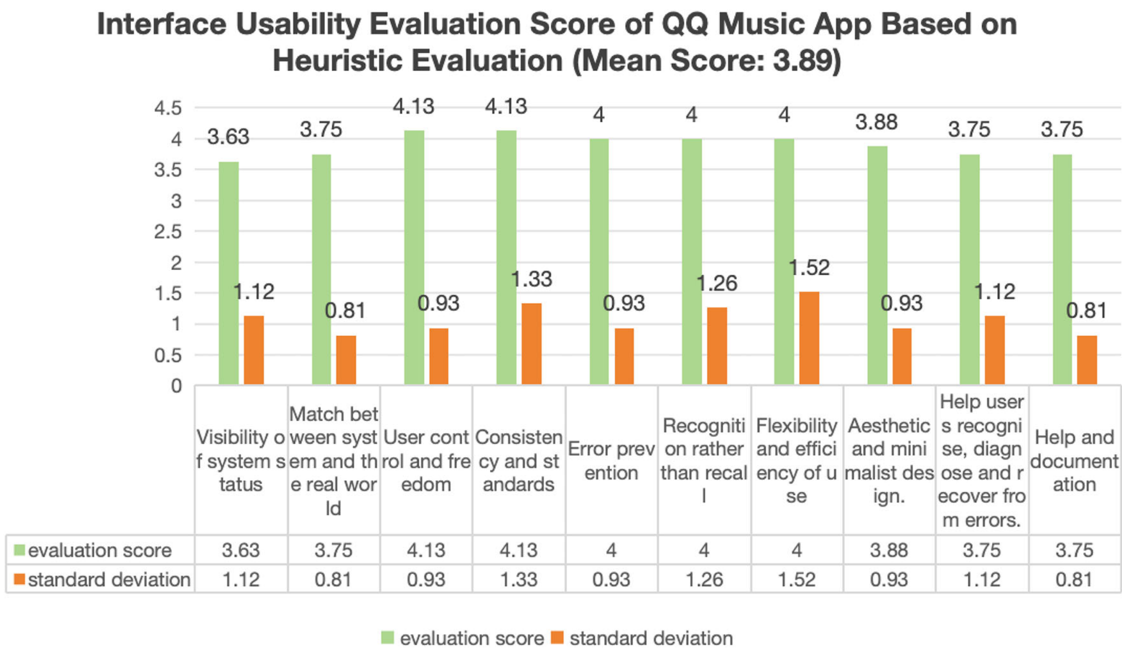

Fig. 4 QQ music app interface usability score based on Jakob Nielsen's usability principles.

As shown in Table 4, the scale results indicate significant usability issues in the QQ Music App interface. Analysis of the severity ratings reveals particularly high scores for several usability concerns: unclear interface visual relationships ($M = 3.60$, $SD = 1.15$), excessive functional information descriptions ($M = 3.47$, $SD = 1.12$), complex content recommendation features ($M = 3.40$, $SD = 1.20$), lack of distinctiveness in function descriptions ($M = 3.38$, $SD = 1.26$), and insufficient interface operation feedback ($M = 3.38$, $SD = 0.93$). These issues demonstrate substantial adverse effects on the overall interface usability of the QQ Music App.

In order to grasp the reasons for the mean score of the QQ Music App interface usability severity scale, a 30–60 min post-test interview was conducted with 15 testers who participated in the heuristic evaluation. As shown in Fig. 5, the main usability problems of the QQ Music App interface were analyzed according to the Usability Problem Severity Scale and combined with the results of the interviews:

First, QQ Music App interface visual relationship and interaction logic. Firstly, QQ Music App has set up functions such as recommendation, e-books, entertainment games and so on in the navigation bar of the interface homepage, without clearly distinguishing the visual priority between core and secondary functions, which easily leads to the confusion of the user's interface operation path and reduces the efficiency of use. Second, the background color of the home screen lacks sufficient contrast with the functional areas. The layout of interface elements—such as text, buttons, and icons—is also crowded, which makes it harder for users to locate features and understand the interface content. Lastly, in terms of interaction, there are no clear visual boundaries between the navigation bar, main menu, and function icons. This lack of separation creates blurred boundaries between clickable areas and functional zones, making it easy for users to accidentally tap on the wrong icon or function. Representative interview data are as follows:

> "The interface navigation bar design is more complicated. The color contrast relationship on the home page is not obvious enough, often causing visual fatigue. The area for clicking icons is not well delineated, making it easy to accidentally touch other functions" (Participant P1, P13)

Second, interface functional layout and information density. The QQ Music App has introduced more than a dozen different music recommendation modes. For example, the homepage features various song recommendations such as"Recommended for You," "Welcome, Feel Free to Listen," "Personalized Radio," and "Hi, Curated for You Today."However, this overabundance of options tends to increase users' decision-making burden. In addition, the high information density of the interface may obscure the songs users truly intend to listen to, thereby distracting their attention and interfering with the effective use of the interface. Representative interview data are as follows:

Table 4 Severity of usability problems.

NO	Interface level	Usability issue description	Violated heuristic principles	Average severity	SD
1	Homepage/Secondary Interface	Unclear interface visual relationships	Interface visibility, learnability, operational efficiency, aesthetic design, consistency and standards	3.60	1.15
2	Homepage	Excessive functional information descriptions	Interface visibility, operational efficiency, interface learnability, consistency and standards	3.47	1.12
3	Homepage/Secondary Interface	Complex content recommendation features	Interface visibility, interface learnability, system and reality match, error prevention	3.40	1.20
4	Homepage/Secondary Interface	Lack of distinctiveness in function descriptions	Interface visibility, interface learnability, operational efficiency	3.38	1.26
5	Homepage/Secondary Interface	Insufficient interface operation feedback	Interface visibility, interface learnability, operational efficiency, interface assistance, error prevention	3.38	0.93
6	Homepage/Secondary Interface	Excessive pop-ups and complex information hierarchy	Operational efficiency, interface visibility, error prevention	3.25	1.20
7	Homepage/Secondary Interface	Inappropriate color composition	Interface visibility, operational efficiency, interface aesthetic design	3.25	1.52

> "The QQ Music App offers too many types of music recommendations, which makes it confusing to use. There are so many recommendation features that sometimes I don't know which one to choose." (Participants P7, P9)

Third, functional icon recognition and interaction consistency. First, the functional descriptions lack sufficient recognizability. For instance, the "Listen to Songs" and "Favorites" functionalities are at the top of the QQ Music App homepage. The interface design solely utilizes icons to communicate functional information, potentially leading to erroneous user operations. Furthermore, the usage of obscure terminology for function labels imposes an extra cognitive strain on the interface. This is due to varying preferences for icon usage among different Internet apps available on the market, as well as cognitive disparities in users' comprehension of icons and functional utilization. The UI exhibits an inconsistent exit mode. The QQ Music App has implemented a downward and leftward page-sliding interface exit mode, which complicates operations and increases operational memory costs. Representative interview data are as follows:

> "The interface function icons lack recognizability, which makes it easy for users to make mistakes in understanding the functions. Inconsistency in the way the interface exits or returns to the operation." (Participant P3, P15)

Fourth, feedback and guidance on the operation of the QQ Music App interface. In the outdoor environment, when the user activates the "Your Song List Treasure Trove" function, the interface solely communicates operational feedback by altering the icon color, lacking additional, clearer feedback, which complicates the user's ability to confirm operations. Furthermore, upon clicking the function, the interface remains on the music playback home page, failing to present a new page mode for song selection, thereby imposing additional burdens on the user regarding song collection, categorization, and cancellation. Representative interview data are as follows:

> "The operation feedback on the interface is not clear enough. When I tap on a song recommendation icon, there's no feedback telling me the action was successful. It's also hard to find the songs or playlists I've already saved." (Participant P3)

Fifth, QQ Music app interface pop-ups and information hierarchy: The QQ Music App has set up "VIP songs for free for 10 min," a "customize player," or various commercial advertisement pop-ups in the interface start-up phase and secondary interface switching phase. These new interactive clicking behaviors are likely to cause a complicated operation process and reduce the smoothness of the interface. In addition, some functions of the QQ Music App have complex information hierarchies. For example, the "Listen to Songs" function has set up shortcut buttons on the song search page and the home page. However, the "Listen to Songs" function is still provided on the secondary page of "Personal Settings," which makes users feel overloaded with information, making it difficult to quickly locate the target function, especially for users who are unfamiliar with the interface, which is more likely to create a sense of confusion and lead to a more complicated operation process. It is easy for users to feel overloaded with information and difficult to quickly locate the target function. Representative interview data are as follows:

> "There are too many ads and pop-up prompts in the interface, which makes it harder for users to use the app. The 'Song Recognition' feature is already on the home screen, but it also appears in the personal center. This kind of repeated function setup can cause confusion during operation." (Participants P9, P15)

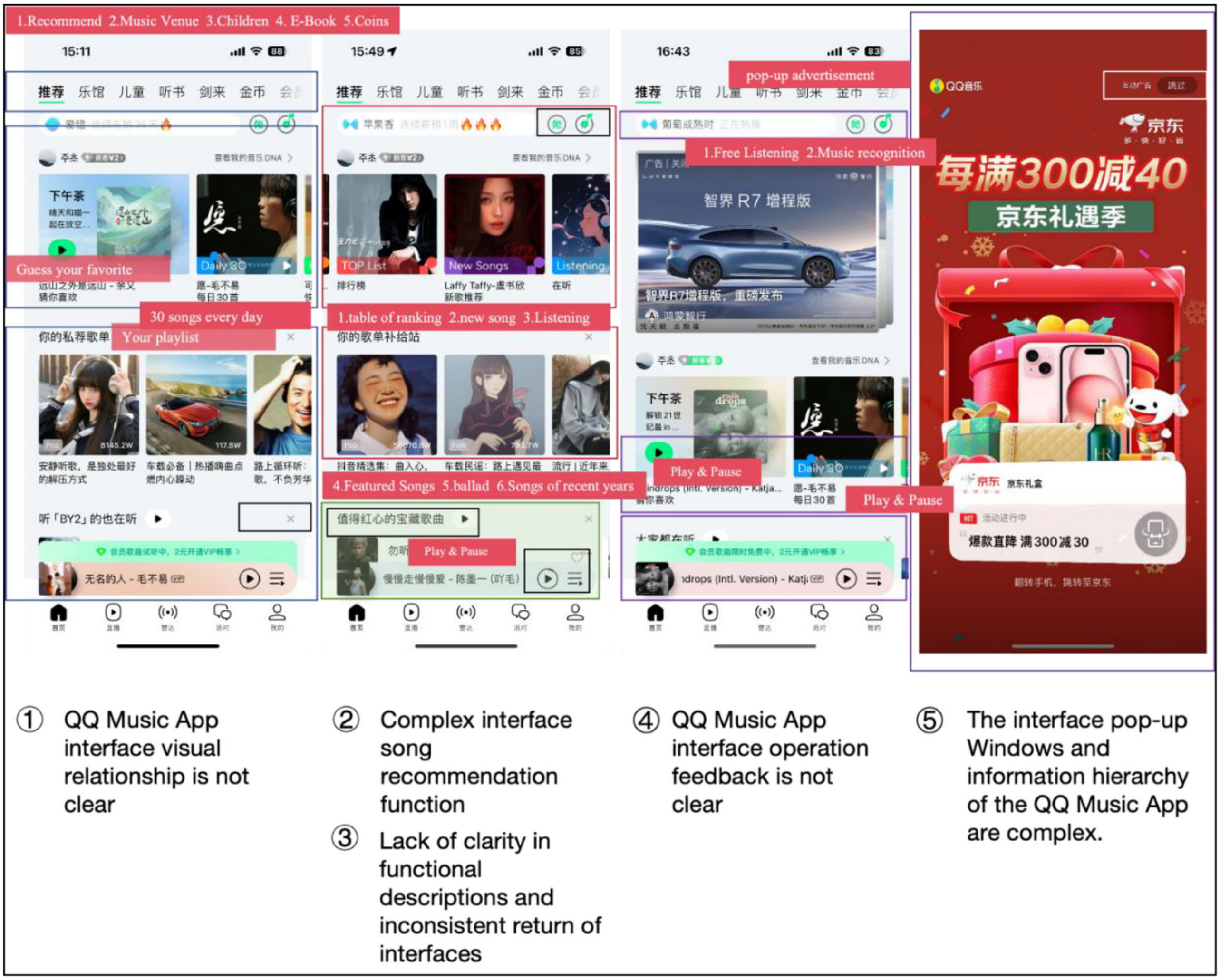

Fig. 5 Usability issues of QQ music application program interface based on heuristic evaluation.

Design strategy and interface prototype

Design strategy. The usability evaluation of the QQ Music App interface yields the following conclusions: Overall, the interface usability of the QQ Music App satisfies the requirements of the majority of users; however, significant usability issues persist, including ambiguous visual relationships, excessive textual descriptions, and function descriptions lacking clarity. The study presents design techniques aimed at enhancing the interface usability of the QQ Music App to offer fresh insights for improving the usability of online music applications.

Initially, enhancing interface color clarity and reducing information overload. In terms of interface color relationships, the QQ Music App can enhance operational efficiency by employing diverse color combination strategies, including the judicious application of color hierarchies, optimization of color contrast, and the creation of a cohesive color system through elements such as hue, size, placement, and contrast. People prefer streamlined and intuitive interface functionalities in today's fast-paced environment. The QQ Music App can enhance its homepage song suggestion feature by prioritizing regularly utilized recommendation services, thus minimizing users' decision-making complexity. Additionally, to augment interface usability, the program may refine icon display modes and enhance feature visibility via hierarchical menus or advanced search functionalities. For online services unfamiliar to users, the interface may include textual and iconic components to highlight the functionality and offer assistance, thus enhancing feature usage and perceptual efficiency.

Secondly, provide interactive prompts for pop-up management. The QQ Music App UI features numerous pop-up windows, potentially increasing users' cognitive stress. Recurrent interface notifications not only heighten the probability of operational errors from inadvertent touches but also augment cognitive load and disrupt operational flow. To further diminish the likelihood of user operational errors, it is essential to establish pop-up management features and introduce a streamlined interface mode. Moreover, we could centralize recurrent advertisement pop-ups in designated interface areas to enhance user concentration during interaction.

Third, executing contextually relevant and transparent operational feedback. Interface feedback must communicate task completion and operational status to users. In modern interface design, relying too much on visual feedback can make users less aware of how things work. This is especially true in interfaces with a lot of functions or complicated settings for interactions, where lack of visual contrast makes icon-based feedback less useful. An in-depth analysis of feedback systems and notification methods is crucial to alleviate user irritation arising from ambiguous operational progress. The interface needs to give different levels of feedback depending on the environment. It

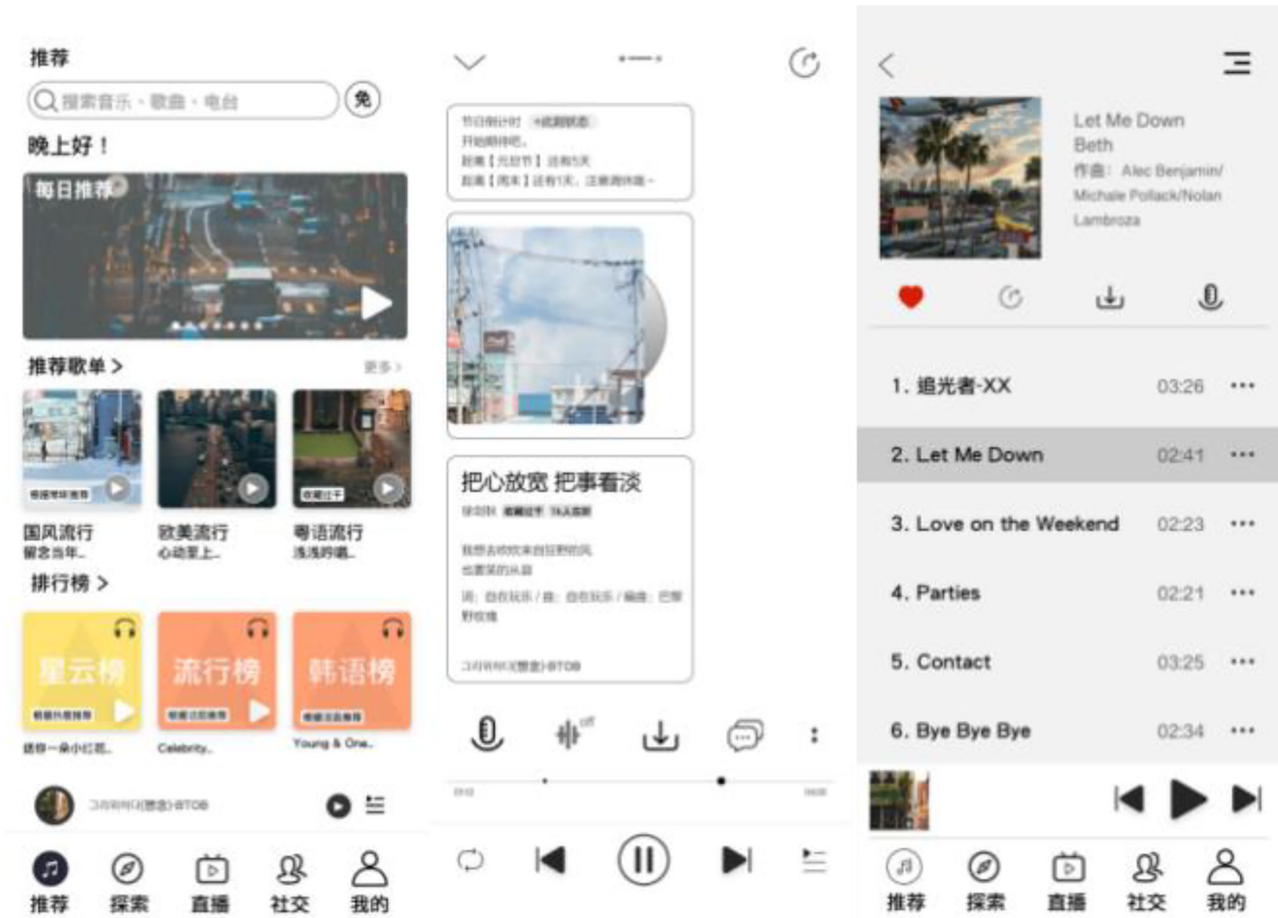

Fig. 6 Home page and its music playback page.

needs to take into account how design elements like layout, icons, contrast ratios, fonts, and functional density interact with each other. Effective feedback must incorporate haptic signals nd auditory signals to convey operating status clearly. Furthermore, separate operating zones for various functional sectors promote swift acclimatization to interaction patterns and improve operational efficiency.

Fourth, alleviating cognitive stress during interface engagement. The QQ Music App interface can be enhanced by prioritizing important information and relegating secondary content to optional or deeper hierarchical levels, hence minimizing the cognitive load on users. Interface layouts must be optimized by consolidating functions and reducing operating options to avoid decision paralysis. The information density in individual interfaces should be reduced, possibly by using common terminology in function names to lessen cognitive demands during feature exploration. The implementation of adjustable interface configurations would allow users to adjust layouts and functionality displays based on personal preferences, including the ability to hide or anchor specific functional modules.

Interface prototype. To further evaluate whether the proposed design strategies could effectively enhance the usability of music apps, this study developed a new-generation music app interface prototype based on the research findings and strategic recommendations. Compared to the previous interface, the improved prototype version incorporated several key interface design enhancements, including a more streamlined interface structure, more intuitive text and icons, and simplified operational procedures. The research team members conducted multiple rounds of design iterations on the interface prototype to ensure the design outcomes would better meet users' interface usability requirements. We developed the final interface prototype based on the basic interface functionalities of the QQ Music App (Figs. 6, 7).

Usability test evaluation. Five target users were recruited to conduct A/B testing on both the existing QQ Music app interface and the improved interface prototype. As part of the evaluation, the study designed common music app operation tasks to assess the interface prototype's usability. To gather subjective feedback, participants completed the System Usability Scale (SUS) questionnaire and engaged in post-test interviews.

From the SUS (System Usability Scale) results, the existing version of the QQ Music app received a SUS score of 69, while the improved interface prototype achieved a SUS score of 75.75 (Table 5). This indicates that the interface prototype modified according to the design strategies enhanced users' efficiency, effectiveness, and satisfaction when using the interface, and received higher usability ratings from participants.

Post-test interview results showed that all participants acknowledged improvements in interface effectiveness due to the streamlined homepage functionality, prioritized display of frequently used features, and clear interface navigation. The improved interface's function labels and text prompts better aligned with users' cognitive habits, effectively enhancing interface efficiency and increasing users' confidence in their operational behaviors. These findings indicate that improving the existing QQ Music App based on the aforementioned research

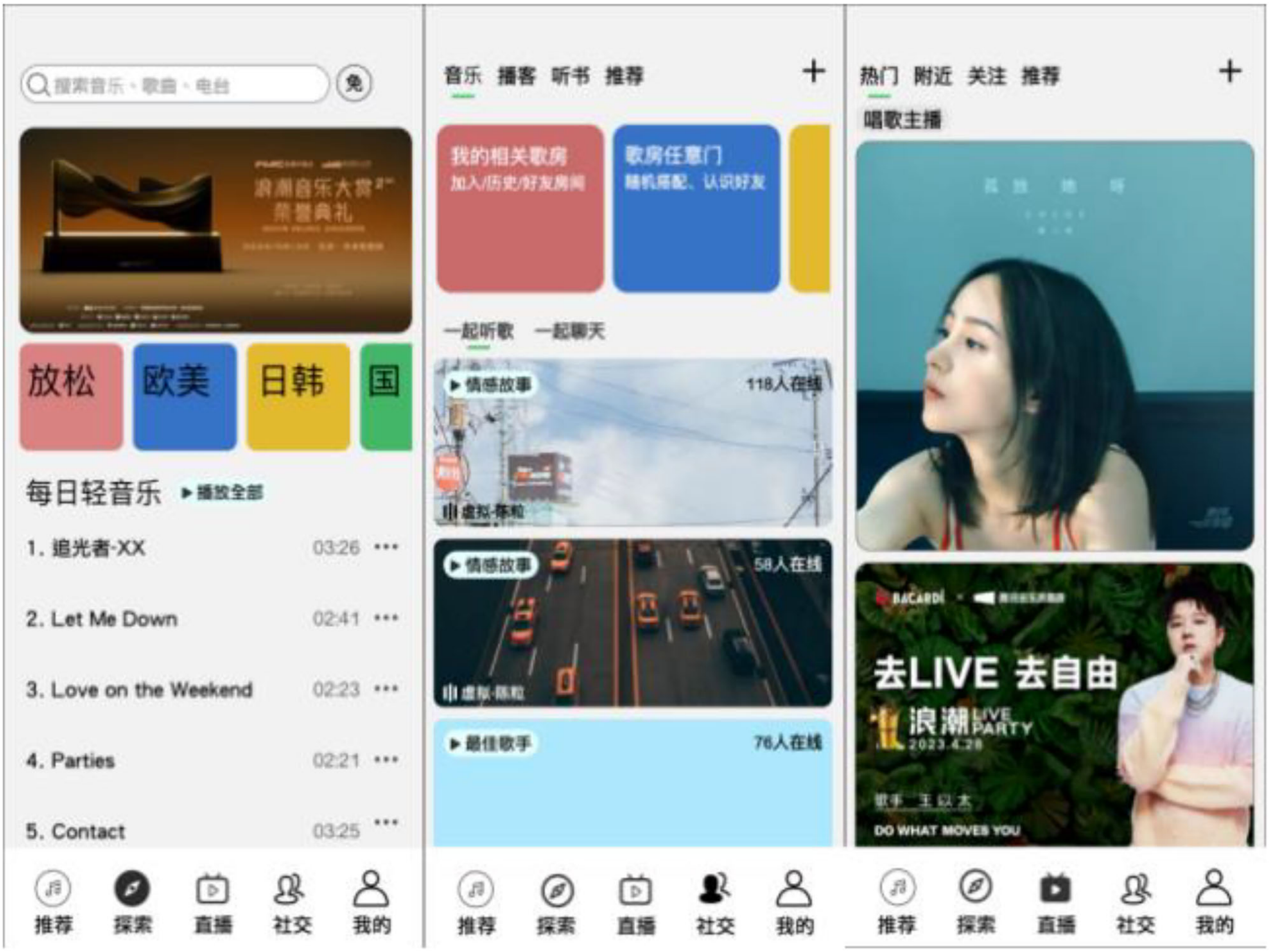

Fig. 7 Other music recommendations page.

strategies can further enhance the usability of music app interfaces for users.

Discussion and conclusions

Discussion. Interface usability is a vital subject in human-computer interaction research pertaining to online music applications. Computer technology's advancement and proliferation intricately link to the emergence of human-computer interaction (Gupta et al., (2020); Harish et al., 2013). Effective human-computer interaction prioritizes the design of computer systems that are secure, efficient, user-friendly, and comprehensive (Vora, 1998; Issa and Isaias, 2022). As technology growth progresses in society, the functionality and engagement methods of modern online music applications are growing more advanced. In light of this tendency, we deem it essential to reassess the usability of interfaces in online music applications. The interface styles of music applications demonstrate unique design and interaction patterns that set them apart from other application categories. The primary purpose of music applications is online music discovery and recommendation. Music applications primarily structure their features to categorize music by genres, styles, artists, and rankings, thereby facilitating individualized song recommendations for consumers. Thus, tailored suggestion features constitute a fundamental attribute of music applications. Moreover, modern online music platforms have evolved to include a broader array of music-related and social functionalities, including audiobooks, social games, and live music streaming. This evolution demonstrates that contemporary music applications are transitioning into internet platforms with social entertainment features.

This study revealed that the existing interface layout, color schemes, and functional configurations of the QQ Music App exhibit significant usability issues, resulting in a heightened cognitive load for users interacting with the interface. Cognitive load indicates the extent of mental effort a user experiences in working memory while utilizing an application (Weichbroth, 2020). The current iteration of the QQ Music App interface offers users between twenty and thirty categories of music recommendations, which complicates their decision-making process due to the increased diversification of alternatives. Furthermore, issues such as interface pop-ups, inadequate contrast between background colors and icons, and other visual interactions within the interface are creating new challenges for users. The study indicates the necessity of simplifying interface operations to diminish users' cognitive burden. It advocates for minimizing operational steps and incorporating familiar design elements to facilitate user learning, ultimately leading to a more user-friendly interface that alleviates cognitive strain during navigation.

This research is considered topical and significant. It provides profound insights into the usability challenges faced by users of online music applications and more intricate interface interactions. The study reveals multiple significant implications. This research contributes to theoretical development by elucidating usability testing as a critical quality trait for assessing user interface efficiency, recognizing that several product categories

Table 5 Usability evaluation results of QQ music interface and improved interface prototype SUS system.

	System usability scale questions	Average scores	Confidence interval	Usability score for design prototypes	Confidence interval
1	I think that I would like to use this system frequently.	3.9	[3.49, 4.31]	4.2	[3.75, 4.65]
2	I found the system unnecessarily complex.	2.6	[2.10, 3.10]	2.2	[1.75, 2.65]
3	I thought the system was easy to use.	3.8	[3.50, 4.10]	4	[3.52, 4.48]
4	I think that I would need the support of a technical person to be able to use this system.	1.9	[1.49, 2.31]	1.6	[1.23, 1.97]
5	I found the various functions in this system were well integrated.	3.6	[3.23, 3.97]	3.8	[3.35, 4.25]
6	I thought there was too much inconsistency in this system.	2.3	[1.82, 2.78]	2	[1.66, 2.34]
7	I would imagine that most people would learn to use this system very quickly.	3.7	[3.11, 4.29]	4.1	[3.68, 4.52]
8	I found the system very cumbersome to use.	2	[1.52, 2.48]	1.9	[1.49, 2.31]
9	I felt very confident using the system.	3.9	[3.33, 4.47]	4.2	[3.9, 4.5]
10	I needed to learn a lot of things before I could get going with this system.	2.5	[2.02, 2.98]	2.3	[1.95, 2.65]
	Total skor	69		75.75	

prioritize different usability metrics. This study is significant as it enhances and deepens our comprehension of usability difficulties across various products, contributing meaningfully to the theoretical development of usability frameworks. Examining usability challenges in online music apps offers novel case studies and empirical evidence for the domain of human-computer interaction, thus enhancing the theoretical framework of human-computer interaction.

This research's practical value is its heuristic evaluation of the QQ Music application's interface usability in the Chinese market, highlighting areas for enhancement in online music application interface usability. This research can aid developers in identifying and addressing usability issues faced by users of online music applications, thereby improving interface usability as these applications evolve with enhanced and diversified functionalities.

Conclusion. This research is based on the contemporary developmental trends of increasingly intricate interface interactions and a broader range of features in online music applications. The objective is to analyze the usability of online music applications and propose enhancements for improved user experience. The research examines QQ Music, a popular music application in China, and uses heuristic evaluation to assess the usability of its interface.

The results show that the current version of the QQ Music interface has a number of usability issues, including unclear visual links between interface elements, poor functional explanations, too many interface pop-ups, and not enough operational feedback. Based on the results of the heuristic evaluation, this study suggests that when designing the interface of future online music apps, the amount of functionality should be planned strategically, commonly used features should be made easier to see, pop-up displays should be customized, and there should be different levels of context-sensitive operational feedback.

Additionally, the study created a novel interface prototype for QQ Music to evaluate the enhancements in usability following the implementation of these design alterations. The revamped QQ Music interface, informed by the suggested design principles, significantly improved user interface usability. The research findings offer novel insights for QQ Music's design and development team concerning enhancements in interface usability.

Research limitations and future research directions. Like all studies, this research has certain limitations. First, the usability evaluation focused solely on online music app interfaces within the Chinese market. As a result, the findings are more applicable to understanding the issues Chinese users face when using such interfaces. Given the differences in political systems, cultures, social contexts, and daily habits across countries, the insights gained from this study have limited applicability to the design of online music app interfaces in other regions. Second, this study involved only 15 domain experts for the heuristic evaluation, which suggests that the sample size remains relatively small. Therefore, in future research, we plan to focus on cross-national usability analysis of online music app interfaces and expand the number of participants involved in heuristic evaluations to enhance the value and generalizability of the findings.

Data availability

This study primarily employed heuristic evaluation, interviews, and post-test questionnaires as research methods. All data generated or analysed during this study are included in this published article.

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Author Contributions

Chao Zhou: conceptualization, research design, data collection, data analysis, and manuscript writing. Hongyuan Zhao: methodology, empirical validation, and review of results.

Competing interests

The authors declare no competing interests.

Ethical approval

This study does not require formal ethical approval, as it does not involve human participants who would need to be reviewed in accordance with institutional or national ethical regulations. According to Article 15, Paragraph 2 of the Korean Bioethics and Safety Act and Article 13 of its implementing regulations, research that does not involve vulnerable populations, invasive procedures, physical interventions, or the collection of personally identifiable or sensitive information is exempt from Institutional Review Board (IRB) approval. Specifically, the law stipulates that research utilizing survey responses or behavioral observations is exempt from formal IRB review as long as participants cannot be individually identified and no sensitive data is collected. In accordance with the aforementioned legal provisions, this study focuses on participants' feedback regarding their usage behavior of the QQ Music App interface and their cognitive understanding of individual behavior. The methods employed include observation, interviews, and questionnaires to collect both subjective and objective information from participants, without collecting personal identity information or implementing interventions that may pose risks to participants. The study does not involve vulnerable populations, and all responses are collected anonymously. Given the nature of the study, no formal review or exemption was sought from the Institutional Review Board. This study adheres to the ethical guidelines of the National Research Foundation of Korea and the principles of the Declaration of Helsinki, ensuring compliance with ethical research standards.

Informed consent

Prior to conducting this study, we provided each participant with a written informed consent form. All participants were fully informed of the purpose and objectives of this study, which involved conducting a heuristic evaluation of the QQ Music App. Participants were informed of the study's purpose, the voluntary nature of their participation, and their right to withdraw at any time without any consequences. The study emphasized confidentiality and data protection, ensuring that all data was anonymized and securely stored. We took measures to ensure that no personally identifiable information was collected and that data was encrypted during transmission and storage to protect participants' privacy. After confirming the anonymity of their participation and that the collected data would be used solely for academic research, they voluntarily signed the informed consent form. The informed consent forms were obtained between February and March 2024.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1057/s41599-025-05424-4.

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